

CoRoT-2b

R-1051

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_151027 · created 2026-07-30T09:45:38+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457322.639 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.002 +/- 0.062
Transit depth (Rp/Rs)^2	0.00025 +/- 0.02 [%]
Orbital Inclination (inc)	88.99 +/- 2.06
Airmass coefficient 1 (a1)	0.84 +/- 0.5
Airmass coefficient 2 (a2)	0.08 +/- 0.13
Scatter in the residuals of the lightcurve fit is	60.5 %
Best Comparison Star	None
Optimal Aperture	1.11
Optimal Annulus	6.5
Transit Duration (day)	0.084 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.002 ± 0.062 vs 0.1667 (deviation 1.68σ)
Duration fit vs published	0.084 d vs 0.095 d (deviation 1.36σ)
Mid-time O-C	-3.1 min
Depth SNR	0.0
Residual scatter	60.5 %

Light curve

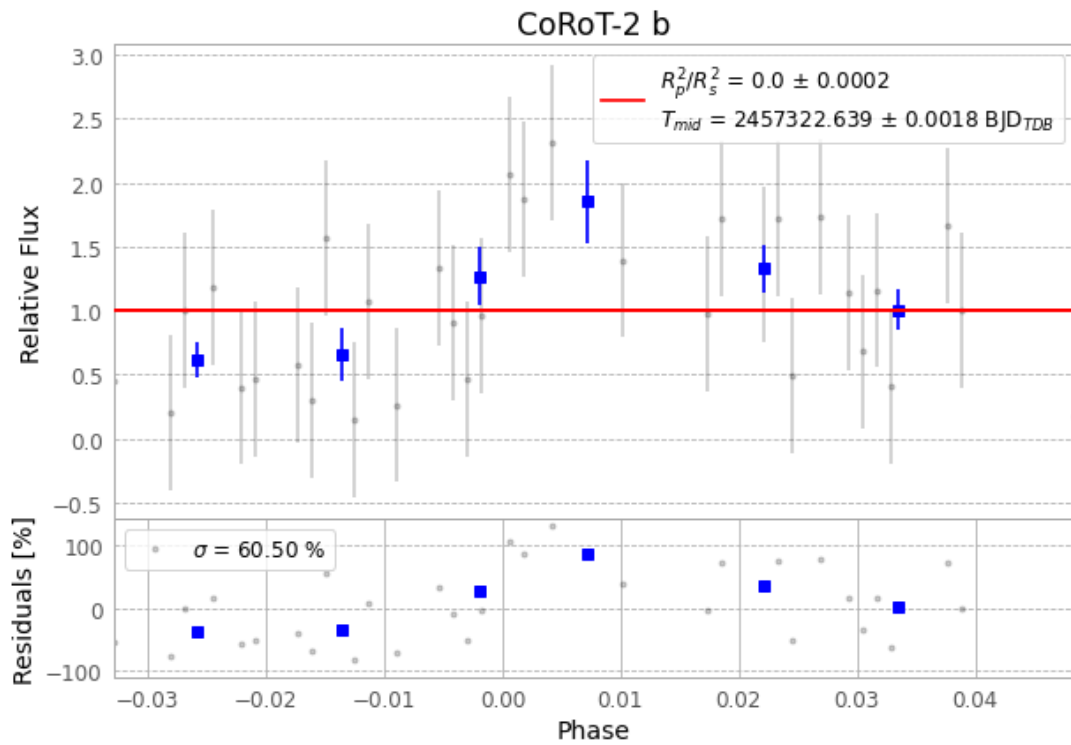


Figure 1 — FinalLightCurve_CoRoT-2 b_2015-10-26.png (SHA-256 1ab3617f54d6513f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 445], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6ccbea654e3959d8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), CoRoT-2151027015712.FITS → CoRoT-2151027052112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-151027.log (f5915b9faedb...), FinalLightCurve_CoRoT-2 b_2015-10-26.png (1ab3617f54d6...), FinalLightCurve_CoRoT-2 b_2015-10-26.csv (30b88b11d5b5...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 09:45Z.